

FAPE PROJECT
PRE-TEST GENERAL EDUCATION (200 ITEMS)

Exam Breakdown

40 questions	20% English
30 questions	15% Filipino
40 questions	20% Mathematics
40 questions	20% Science
40 questions	20% Social Sciences
10 questions	5% Information and Communication Technology

ENGLISH (40 points)

For numbers 1-10, select the letter that corresponds to the BEST verb for each sentence. Observe the rules of the subject-verb agreement.

1. Margarita is the only one of those students who _____ lived up to the dreams stated in their high school yearbook.
A. has
B. have
C. is
D. are
2. The Rotary Club, as well as the Lions Club, _____ to submit a new constitution.
A. need
B. needs
C. will needs
D. is need
3. One of my best friends _____ an extra on *Mara Clara* this week.
A. are
B. was
C. will
D. is
4. Not only the students but also their instructor _____ been called to the principal's office.
A. have
B. has
C. are
D. is
5. Most of the cow's milk _____ gone stale. Three gallons of milk _____ still in the refrigerator.
A. has, are
B. have, is
C. has, is
D. have, are

6. Each and every student and teacher in this university _____ for much needed changes by next year.
A. will hopes
B. is hopes
C. hope
D. hopes
7. The students and faculty _____ for a new College of Medicine building by next school year.
A. hopes
B. hope
C. will hopes
D. is hope
8. Rice and fish, my favorite meal, _____ me of my native country, the Philippines.
A. remind
B. reminds
C. reminding
D. will remind
9. A large number of voters still _____ uneducated voters, who are swayed by money, power or influence.
A. remain
B. remains
C. will remained
D. remaining
10. Many a man _____ a great woman behind his success.
A. are
B. is
C. has
D. have

For numbers 11-20, select the most APPROPRIATE word for each sentence. Observe conventions for proper usage in the English language.

11. The rich family _____ the homeless boy.
A. adapted
B. adopted
C. adaptive
D. adoptable
12. _____ technology will be used in this war.
A. Advanced
B. Advance
C. Advancing
D. Advanse
13. The bride walked down the church _____ to the tune of wedding bells.

- A. isle
- B. island
- C. aisle**
- D. islet

14. As soon as I put my coat on, I'll be _____.

- A. already
- B. allready
- C. al ready
- D. all ready**

15. I bought shoes at the _____.

- A. bazaar**
- B. bizarre
- C. bazzaar
- D. bazarre

16. I picked up some _____ rocks from the beach.

- A. course
- B. corse
- C. curse
- D. coarse**

17. Any _____ person should know that it is wrong to steal from a bank.

- A. decent**
- B. dissent
- C. descent
- D. descend

18. King Mufasa was trampled by a _____ of wildebeests.

- A. hoard
- B. hoarder
- C. hord
- D. horde**

19. We should be guided by the _____ of truth, fairness and justice.

- A. ideas
- B. ideation
- C. ideals**
- D. idols

20. The decaying corpse is _____ to the crowd.

- A. nauseous**
- B. nauseating
- C. nausea
- D. nauseousating

For numbers 21-25, read the literary piece below. Select the letter that corresponds to the BEST answer for the questions that follow.

EXCERPT FROM THE PORTUGUESE
By Elizabeth Barrett Browning

How do I love thee? Let me count the ways.
I love thee to the depth and breadth and height
My soul can reach, when feeling out of sight
For the ends of Being and ideal Grace.
I love thee to the level of everyday's
Most quiet need, by sun and candle-light.
I love thee freely, as men strive for Right;
I love thee purely, as they turn from Praise.
I love thee with the passion put to use
In my old griefs, and with my childhood's faith.
I love thee with a love I seemed to lose
With my lost saints!--I love thee with the breath,
Smiles, tears, of all my life!--and, if God choose,
I shall but love thee better after death.

21. What type of literary piece is illustrated by the selection?

- A. Sonnet**
- B. Cinquain
- C. Haiku
- D. Ode

22. To whom does the author most likely pertain to, when she uses the word thee?

- A. her dog
- B. her husband**
- C. her mother
- D. her enemy

23. What is the predominant theme of this literary piece?

- A. intense hatred
- B. friendship
- C. enmity and discord
- D. intense love**

24. Which type of figure of speech is predominantly employed in this selection?

- A. hyperbole
- B. onomatopoeia
- C. anaphora**
- D. metonymy

25. What figure of speech is used in the underlined portion of the selection?

- A. alliteration**
- B. assonance
- C. simile
- D. onomatopoeia

For numbers 26-30, read the literary piece below. Select the letter that corresponds to the BEST answer for the questions that follow.

FROM THE MOST DANGEROUS GAME
by Richard Connell

Due to accidental circumstances, Sanger Rainsford (a vicious game hunter) is shipwrecked and stranded on General Zaroff's island. Zaroff's island, also known as ship-trap island, is shrouded in mystery and intrigue. In that island, there is only one superior power, General Zaroff.

Zaroff exhibits callousness toward his favorite prey. In Zaroff's case, the prey is human. Shipwrecked people provide him a constant supply of game. Shocked, Rainsford expresses moral indignation at the general's murderous pastime. Zaroff counters that Rainsford will change his mind when he participates in a hunt. But what Rainsford does not immediately realize is that he will be the quarry.

After Zaroff releases Rainsford into the jungle the next day, Rainsford becomes like the animals he hunts, mere game. After Rainsford sets traps that kill one of Zaroff's tracking dogs and Zaroff's gigantic sidekick, Ivan, he escapes his pursuers by jumping into the ocean from a precipice. That evening when Zaroff goes to bed, Rainsford comes from behind a curtain and confronts the general, saying, "I am still a beast at bay." And the beast then kills the hunter and sleeps soundly in his bed.

26. What is the predominant theme in this selection?

- A. cruelty to animals
- B. inhumanity**
- C. unjust vexation
- D. murder

27. What does the word game pertain to in the underlined portion of this selection?

- A. humans who become Zaroff's prey**
- B. vicious and elusive animals in Zaroff's island
- C. sports like basketball, football and baseball
- D. competition between a hunter and prey

28. How did Rainsford eliminate General Zaroff's dogs and sidekick from the hunt?

- A. by hiding in the jungle
- B. by jumping into the ocean
- C. by killing them with traps of his own**
- D. by paying them with money and food products

29. What do you think happened at the end of the selection?

- A. Zaroff killed Rainsford inside his bedroom.
- B. Rainsford killed Zaroff in a duel.**
- C. Rainsford surrendered to General Zaroff.
- D. Zaroff slept soundly in his bed.

30. Based on this selection, what do you think is "most dangerous game"?

- A. Killing highly dangerous animals
- B. Setting deadly traps in the jungle
- C. Playing the role of God**
- D. Surviving in a dangerous island

For numbers 31-35, read the selection below. Select the letter that corresponds to the BEST answer for the questions that follow.

Ludvig van Beethoven initially dedicated his third symphony to Napoleon, believing that the general intended to sustain the democratic and republican ideals of the French Revolution. However, in 1804, when Napoleon's imperial ambitions became clear, Beethoven took hold of the title page and scratched the name Bonaparte out so violently that he made a hole in the paper. He later changed the work's title to "Sinfonia Eroica" and rededicated it to his patron, Prince Joseph Franz von Lobkowitz, at whose palace it was first performed.

The fourth movement of his Ninth Symphony features an elaborate choral setting of Schiller's Ode to Joy, an optimistic hymn championing the brotherhood of humanity. Since 1972, an orchestral version of this part of the fourth movement, arranged by the conductor Herbert von Karajan, has been the European anthem as announced by the Council of Europe. In 1985 it was adopted as the anthem of the European Community.

31. What does the word imperial refer to?

- A. a crown.
- B. empire.**
- C. aggression against neighbors.
- D. high and mighty.

32. Why did Beethoven drop Napoleon from the title page of the third symphony?

- A. Napoleon's imperial ambitions.**
- B. Napoleon's death.
- C. arguments with Napoleon.
- D. Napoleon's lack of interest.

33. For which of the following was the Ode to Joy adopted?

- A. as Napoleon's anthem.
- B. as the hymn of Prince Lobkowitz's palace.
- C. as Germany's anthem.
- D. as the anthem of the European Community.**

34. What does the word republican mean in the context of this selection?

- A. members of the Republican Party.
- B. members of the Senate and House.
- C. that the supreme power of government lies with voting citizens.**
- D. anarchists.

35. Who was the author of the Ode to Joy?

- A. Beethoven
- B. Karajan
- C. Napoleon
- D. Schiller**

For numbers 36-40, read the selection below. Select the letter that corresponds to the BEST answer for the questions that follow.

Thomas Alva Edison developed many devices that greatly influenced life around the world, including the phonograph and the long-lasting, practical electric light bulb. He is also credited with the creation of the first industrial research laboratory.

He held 1,093 U.S. patents as well as many patents in other countries.

His advanced work in telecommunications was an outgrowth of his early career as a telegraph operator.

He originated electric-power generation and distribution to homes, businesses, and factories. His first power plant was in New York City, where in an instant electric street lights were turned on in a section of the city, replacing gas lights.

36. What is Thomas Edison known for?

- A. the electric light bulb only.
- B. many inventions.**
- C. the phonograph only.
- D. telecommunications inventions only.

37. Which among the following best describes Edison's early career?

- A. a telegraph operator.**
- B. a bicycle salesman.
- C. a union organizer.
- D. an insurance executive.

38. What does the Greek word origin of tele (eg, *tele*communications) mean?

- A. hearing.
- B. speaking.
- C. modern.
- D. at a distance.**

39. What does the Latin word origin of generation mean?

- A. enjoy as a family.
- B. make new.
- C. bring forth.**
- D. produce wealth.

40. What older technology did Edison's light bulbs espouse?

- A. gas bulbs**
- B. tanks
- C. atomic bombs
- D. cellular phones

FILIPINO (30 points)

Para sa bilang 41-50, piliin ang titik ng pinaka-angkop na pasawikaing panghalili sa mga salitang may salungguhit.

41. Nadatnan naming naghihingalo sa ospital ang batang nabundol ng kotse sa Commonwealth.
- A. agaw-buhay**
 - B. buntong-hininga
 - C. balat-sibuyas
 - D. kuskos-balungos
42. Mag-ingat sa mga taong iyong pinagkakatiwalaan. Ang iba sa kanila'y nagbabait-baitan lamang.
- A. nagbubuhay ng sariling bangko
 - B. balat-sibuyas
 - C. buto't balat
 - D. bantay-salakay**
43. Walang pakundangan ang pagpatay ng mga tulisan sa mga taumbayan.
- A. nagdilim ang paningin
 - B. halang ang kaluluwa**
 - C. matigas ang buto
 - D. hampas-lupa
44. Tampalasan ang taong nagnakaw ng aking pagkain kaninang tanghali.
- A. nagdilang-anghel
 - B. nagdilang-demonyo
 - C. maitim ang budhi**
 - D. makalaglag-matsing
45. Hindi mabuti ang pagkakalat ng tsismis sa ating paaralan.
- A. kaututang dila**
 - B. maanghang and dila
 - C. kuskos-balungos
 - D. daga sa dibdib
46. Ang pagsisinungaling ay kasalanan sa Diyos at sa kapwa.
- A. pagdidilang-demonyo
 - B. sanga-sangang dila**
 - C. may krus ang dila
 - D. nagdilang-anghel
47. Walang magandang dilag sa kanilang paaralan na nakakalusot sa paningin ni Miguel.
- A. tatlo ang mata
 - B. namuti ang mata
 - C. matulis ang mata
 - D. matalas ang mata**
48. Ang taong batugan ay kadalasang walang trabaho. Wag natin silang tularan.
- A. makitid ang utak
 - B. nagpupusa
 - C. nagbibilang ng poste**

D. ningas-kugon

49. Pilit na sabay niligawan ni Marco sina Aida at Fe. Siya ay isang salawahan.

- A. naniningalang-pugad
- B. namamangka sa dalawang ilog**
- C. nagmumurang kamatis
- D. sampid-bakod

50. Hindi makabuo ng isang matibay na paniniwala si Jane. Pabago-bago palagi ang kanyang pananaw.

- A. talusaling
- B. takipsilim
- C. talusira**
- D. tawang-aso

Para sa bilang 51-60, piliin ang titik ng pinaka-angkop na salita para sa bawat pangungusap.

51. Bakit ka tawa _____ tawa?

- A. ng
- B. nang**
- C. upang
- D. sa

52. _____ tayo sa swimming pool. Kay sarap magtampisaw at lumangoy.

- A. Lumusong**
- B. Sumuong
- C. Umahon
- D. Humango

53. Kapag kumulo na yung sabaw, maaari mo na itong _____.

- A. isalang
- B. isalok
- C. hanguin**
- D. maghango

54. Sa kanto ng Pedro Gil tayo _____ ng bus papuntang Fairview.

- A. mag-sakay
- B. mag-bantayan
- C. magpunta
- D. mag-abang**

55. Halos lahat ng palabas sa telebisyon ay _____ sa kasalan nina Prince William at Kate Middleton.

- A. tungkol**
- B. ukol
- C. para
- D. upang

56. _____ pa rin nila ang nawawalang singsing ni Monica.

- A. Nakikita
- B. Naghahanap
- C. Humahanap
- D. Hinahanap**

57. Wala siyang ginawa _____ matulog at kumain. Siya ay nagpapaka-batugan.

- A. kung di
- B. kung hindi
- C. kundi**
- D. kungdi

58. Tinangay _____ magnanakaw na nakasakay sa motorsiklo ang aking bag.

- A. ng**
- B. nang
- C. nung
- D. ni

59. _____ nila kami ng maraming gawain.

- A. Iniwan
- B. Iniiwan
- C. Iniwanan**
- D. Pinag-iwan

60. Dahil wala ka namang _____ sa akin, huwag mo akong _____ ng anumang pasalubong.

- A. ipinadala, hanapan**
- B. pinadala, hanapin
- C. ipinagpadala, maghanap
- D. ipinadala, paghanapin

Para sa bilang 61-65, basahin ang seleksyon. Piliin ang titik ng tamang sagot sa mga katanungang sumusunod.

Sa mga panhong sunud-sunod ang dagok na dumating sa buhay ng tao, animo'y madilim ang kapaligiran, parang gabi. Subalit sa kabila ng kadiliman, darating at darating din ang sandaling sisikat ang araw at kasabay ng liwanag na ito ay isang magandang pag-asa sa buhay ang kakaharapin. Ganyan ang buhay: kung may dilim, may liwanag; kung may gabi ay may araw; parang gulong: minsang nasa ibabaw, minsan ay nasa ilalim.

61. Anung uri ng pamamahayag ang ipinamalas sa seleksyon?

- A. Paglalahad**
- B. Paglalarawan
- C. Pagsasalaysay
- D. Pangangatuwiran

62. Paano binigyang-katuturan ng may-akda ang paksa ng seleksyon?

- A. Paghahambing at pagtatambis**
- B. Paglalarawan
- C. Paghahalimbawa

D. Pagkilala sa sanhi at bunga

63. Ano ang ibig sabihin ng, “sunud-sunod na dagok?”

- A. kamatayan
- B. pagkakasakit ng kanser
- C. maraming problema**
- D. pagkakabugbog

64. Alin sa mga sumusunod ang mensahe ng may-akda sa mambabasa?

- A. Huwag mawalan ng pag-asa.**
- B. Magpatiwakal kung ikaw ay nalulungkot.
- C. Panandalian lamang ang ligaya sa buhay.
- D. Magtiwala sa Poong Maykapal.

65. Ano ang ibig sabihin ng, “sisikat ang araw?”

- A. mag-uumaga na
- B. darating ang mabuting kapalaran**
- C. mabubuksan ang isipan sa katotohanan
- D. mananalò sa megalotto

Para sa bilang 66-70, basahin ang seleksyon. Piliin ang titik ng tamang sagot sa mga katanungang sumusunod.

Mula sa Isang Dipang Langit
ni Amado V. Hernandez

Saknong 3:

Sa munting dungawan, tanging abot-malas
ay sandipang langit na puno ng luha,
maramot na birang ng pusong may sugat,
watawat ng aking pagkapariwara.

Saknong 4:

Sintalim ng kidlat ang mata ng tanod,
sa pintong may susi't walang makalapit;
sigaw ng bilanggo sa katabing moog,
anaki'y atungal ng hayop sa yungib.

Saknong 5:

Ang maghapo'y tila isang tanikala
na kala-kaladkad ng paang madugo
ang buong magdamag ay kulambong luksa
ng kabaong waring lungga ng bilanggo.

Saknong 7:

Kung minsan, ang gabi'y biglang magulantang
sa hudyat - may takas! - at asod ng punlo;
kung minsa'y tumangis ang lumang batingaw,

sa bitayang moog, may naghihingalo.

66.Saan isinulat ng may-akda ang seleksyon?

- A. Sa loob ng bilangguan**
- B. Sa Luneta
- C. Sa isang kampong militar
- D. Sa palengke

67.Anung uri ng tayutay ang ipinapahiwatig sa ikalawang linya ng Saknong 3?

- A. Pagtutulad
- B. Pagwawangis
- C. Pagmamalabis**
- D. Pagsasatao

68.Anung uri ng tayutay ang ipinapahiwatig sa unang linya ng Saknong 4?

- A. Pagtutulad**
- B. Kabalintunaan
- C. Pagmamalabis
- D. Pagpapalit-saklaw

69.Anung uri ng tayutay ang ipinapahiwatig sa ikatlong linya ng Saknong 5?

- A. Pagmamalabis
- B. Pagwawangis**
- C. Pagpapalit-saklaw
- D. Pagsasatao

70.Anung uri ng tayutay ang ipinapahiwatig sa unang linya ng Saknong 7?

- A. Pagmamalabis
- B. Pagwawangis
- C. Pagpapalit-saklaw
- D. Pagsasatao**

MATHEMATICS (40 points)

71.Evaluate the following mathematical expression: $\frac{1+3(5-2 \times 7)+12(9 \times 2)}{1+(-4)-(\frac{80}{5})}$

- A. 11
- B. -12
- C. -10**
- D. 15

72.Which among the following is NOT a prime number?

- A. 31
- B. 41
- C. 71
- D. 91**

73.What are the prime factors of 156?

- A. $1 \times 2 \times 8$
- B. $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$
- C. $2 \times 2 \times 3 \times 13$**
- D. $2 \times 3 \times 3 \times 13$

74. Brenda is buying hotdogs for a picnic. Hotdogs are sold in packages of 10. Hotdog buns are sold in packages of 8. What is the smallest number of hotdogs and buns she can buy to have an equal number of each?

- A. 16
- B. 24
- C. 60
- D. 80**

75. Paz is packaging up presents to give to her friends. She has 45 pencils and 36 stickers. All of the packages need to contain the same amounts of stickers and the same amount of pencils. What is the greatest number of packages that she can make?

- A. 4
- B. 5
- C. 9**
- D. 12

76. The number, 5121152739991320, is NOT DIVISIBLE by which of the following factors?

- A. 3
- B. 4
- C. 7**
- D. 8

77. Two out of every five households have a washing machine. If there are 12,000 households in a certain town, how many do NOT have a washing machine?

- A. 4,800
- B. 5,000
- C. 7,000
- D. 7,200**

78. Monica answered 30% of the 80 items correctly in a math test. How many items did she answer wrong?

- A. 20
- B. 24
- C. 56**
- D. 60

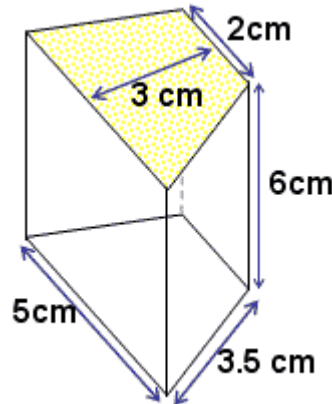
79. A tin can has a radius of 6 cm and a height of 20 cm. What is the volume of milk that this container can hold?

- A. $3,000 \text{ cm}^3$
- B. $2,261 \text{ cm}^3$**
- C. $1,689 \text{ cm}^3$
- D. $2,200 \text{ cm}^3$

80. The total surface area of a cubic box is 1200 cm^2 . What is the length of one side of this box?

- A. 11 cm
- B. 12 cm
- C. 13 cm
- D. 14 cm**

81. What is the total surface area of the trapezoid prism shown below?



- A. 75 cm^2
- B. 85 cm^2
- C. 105 cm^2**
- D. 115 cm^2

82. Rectangular box A has dimensions 2 cm x 3 cm x 4 cm. Rectangular box B has dimensions 6 cm x 4 cm x 4 cm. What is the ratio of the volume of box A with respect to box B?

- A. 1:2
- B. 1:4**
- C. 2:3
- D. 3:4

83. My school is 25 meters away from my home. How far away is my school from my home in kilometers?

- A. 0.25 km
- B. 0.025 km**
- C. 2,500 km
- D. 25,000 km

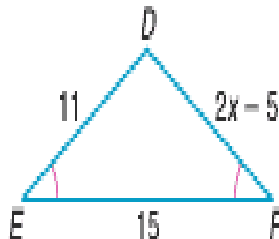
84. A new car measures 1175 kg in weight. How much does it weigh in pounds?

- A. 1,500 lbs
- B. 2,432 lbs
- C. 2,585 lbs**
- D. 3,475 lbs

85. What is the measure of each angle of an equilateral triangle?

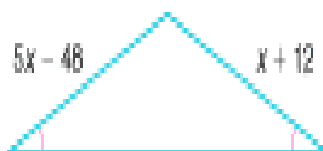
- A. 90°
- B. 30°
- C. 45°
- D. 60°**

86. Triangle EDF is an isosceles triangle. Find the value of x .



- A. 3
- B. 8**
- C. 9
- D. 10

87. What is the value of each of the two sides opposite the congruent angles in the triangle pictured?

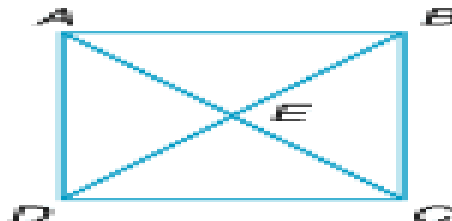


- A. 15**
- B. 39
- C. 27
- D. 75

88. Select the phrase that makes the sentence FALSE. An equiangular triangle _____.

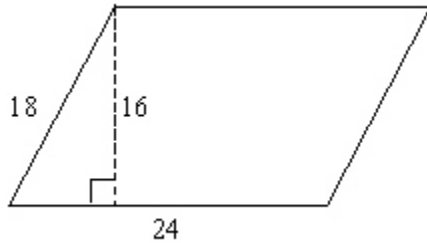
- A. is always an equilateral triangle
- B. is congruent to every other equiangular triangle**
- C. is always an isosceles triangle
- D. is never scalene

89. In rectangle ABCD, $AC = 24$ and $DE = 2x - 8$. Find x .



- A. 16
- B. 13
- C. 10**
- D. 7

90. What is the area of the parallelogram in the illustration below?

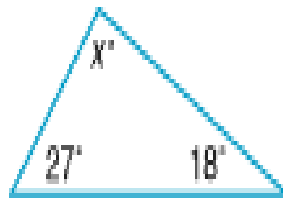


- A. 432
- B. 84
- C. 384**
- D. 192

91. Which statement is NOT true?

- A. A triangle cannot be scalene and isosceles.
- B. A triangle cannot be obtuse and contain a 90° angle.
- C. In an isosceles triangle, the base is congruent to one of the legs.**
- D. A triangle can be obtuse and isosceles.

92. Find the value of x .



- A. 125
- B. 135**
- C. 45
- D. 145

93. A trapezoid has bases of lengths 16 and 28. What is the measure of the midsegment of the trapezoid?

- A. 22
- B. 8**
- C. 14
- D. 17

94. If a parallelogram is equiangular, then the parallelogram can also be considered as what type of quadrilateral?

- A. Rhombus
- B. Rectangle**
- C. Trapezoid
- D. Trapezium

95. What is the solution set (x, y) in system of equations: $2x - 3y = 12$ and $3x + y = 7$?

- A. $(-3, 2)$
- B. $(3, -2)$**
- C. $(2, -3)$

D. $(-2, 3)$

96. A man is thrice as old as his son. In five years, he will be three years more than twice his son's age. How many years old is the man now?

- A. 10
- B. 24**
- C. 41
- D. 36

97. A taxi driver must complete a 180 mile trip in 4 hours. If he averages 50 miles an hour for the first three hours of his trip, how fast, in miles per hour, must he travel in the final hour?

- A. 45
- B. 35
- C. 40
- D. 30**

98. What is the equation of the line that passing through $(1,1)$ and parallel to line $y = 2x - 5$?

- A. $y = 2x + 1$
- B. $y = -(1/2)x + 1$
- C. $y = 2x$
- D. $y = 2x - 1$**

99. Bruno raises a number of ducks and carabaos in his farm, all of which are normal. If his animals have a total of 44 feet and a total of 16 heads, how many carabaos does Bruno have?

- A. 8
- B. 6**
- C. 5
- D. 7

100. A tank can be filled in 6 hours by pipe A running alone, or in 4 hours by pipe B alone. How many hours would be needed to fill the tank if both pipes were running?

- A. $5/12$
- B. $2 \text{ and } 2/5$**
- C. $3 \text{ and } 1/2$
- D. 5

101. The attendance at Enchanted Kingdom on a certain day was 737 persons. If there were 289 more adults than children, how many children were there?

- A. 548
- B. 513
- C. 250
- D. 224**

102. For which value of p in the division $(2x^2 + 11x - p) \div (2x - 3)$ is the remainder zero?

- A. 3

- B. 18
- C. 5
- D. 21**

103. Grace can finish cleaning their room in 45 minutes. When her sister Abby helps her, it takes them 18 minutes to clean the room completely. How many minutes would it take Abby to clean their room alone?
- A. 20
 - B. 30**
 - C. 35
 - D. 25
104. Which among the following is a perfect square trinomial?
- A. $x^2+10x+36$
 - B. $4x^2+10x+25$
 - C. $4x^2-20x+25$**
 - D. $9x^2-12x-4$
105. One marble jar has several different colored marbles inside of it. It has 1 red, 2 green, 4 blue, and 8 yellow marbles. All the marbles are the same size and shape. If Peter takes out a marble from the jar without looking, what is the probability that he will NOT choose a yellow marble.
- A. $8/15$
 - B. $7/15$**
 - C. $7/8$
 - D. $8/7$
106. What is the median of the following set of numbers? {12,8,13,4,7,6,3,3,15}
- A. 12
 - B. 8
 - C. 7**
 - D. 7.5
107. What is the relationship between the mean, the mode and the median for a standard normal distribution?
- A. Mean < median < mode
 - B. Mean = mode = median**
 - C. Mode < median < mean
 - D. Median < mode < mean
108. The alternative name(s) for the median is (are) ____.
- A. 50th percentile
 - B. second quartile
 - C. fifth decile
 - D. all of the above**
109. Two six-sided dice are rolled. Find the probability the first die came up a 2 and the second die came up a 5.
- A. $1/9$
 - B. $1/18$

C. 1/36

D. 1/12

110. Let's say you and your friend play a game that involves several coin flips. You flip three coins at the same time. If all three coins come up the same (all heads or all tails), your friend wins. Otherwise, you win. What is the probability that you win?

A. 3/8

B. 3/4

C. 1/2

D. 1/4

SCIENCE (40 points)

111. Carlo is always interested in asking questions. He is also very eager to find the answers to his questions. What scientific attitude is he displaying?

A. curiosity

B. rationality

C. absence of prejudice

D. wonder

112. Maria respects the ideas of her colleagues even if they are very much contrary to her own beliefs. What important scientific attitude is seen in her actions?

A. honesty

B. absence of prejudice

C. open-mindedness

D. acceptance

113. Pedro weighs evidences dutifully before making any conclusions or decisions. He dissects the information individually before coming to a final conclusion. What scientific attitude is he displaying?

A. carefulness

B. rationality

C. precision

D. accuracy

114. While performing an experiment, Miguel expresses his opinions and tries new ideas even at the risk of failure or criticism. What scientific trait is displayed in this situation?

A. curiosity

B. risk-taking

C. diligence

D. absence of ambiguity

115. Peter admits that he is not free from committing errors. He recognizes that there may be better ideas and realizes that there are individuals whom he may have to consult to arrive at correct observations and conclusions. What scientific trait is he displaying?

A. open-mindedness

- B. objectivity
- C. rationality
- D. humility**

116. While performing research for a pharmaceutical company, Julia gave a truthful report of observations. She did not withhold important information just to please herself or her company. What scientific attitude did she display?
- A. absence of bias
 - B. faith in God
 - C. intellectual honesty**
 - D. critical mindedness
117. What type of reproduction involves gametes?
- A. sexual**
 - B. asexual
 - C. fusion
 - D. fission
118. A new insecticide was very effective in killing flies and mosquitoes. After several months, fewer insects were observed to die from the spray. The reason why fewer insects are killed is that _____.
- A. Insects which survive the spraying developed a liking for the insecticide.
 - B. There is permanent mutation giving insects' resistance to the insecticide.
 - C. Insects which survive the spraying have developed immunity to the insecticide.**
 - D. The insecticide's effectiveness was reduced with aging
119. What stage in the meiosis do the paired homologous chromosomes line up along the equatorial plate?
- A. anaphase
 - B. telophase
 - C. metaphase**
 - D. prophase
120. Bats are classified as mammals because they have _____.
- A. teeth
 - B. feathers
 - C. beaks
 - D. mammary gland**
121. Which of the following does not happen during mitosis?
- A. Coiling up of chromatin
 - B. Replication of DNA strands
 - C. Movement of the chromosomes towards the center
 - D. None of the above**
122. What best describes a cold blooded animal?
- A. the blood is cold
 - B. thrive in cold environment
 - C. body fluid temperature adapts to environment**

D. first order consumers

123. In time of severe cold, animals may hibernate for days or weeks at a time. Their body temperature drops, and they breathe only a few times in a minute. A hibernating animal uses the fat store in its body for nourishment. In times of severe heat, other animals may estivate. Just as with hibernating animals, the body function of estivating animals slow down almost to a stop. To an observer, the animal appears to be either in deep sleep or dead. What generalization could be accurately said about hibernation and estivation?

A. Both are responses to extreme temperature.

B. Both use more energy than when the animal is awake.

C. Both are responses to summer.

D. Both are responses to winter.

124. The plasma membrane is composed of two layers of _____ interspersed with cholesterol and proteins.

A. hydrophilic molecules

B. phospholipids molecules

C. hydrophobic molecules

D. lipid molecules

125. What symbiotic relationship refers to the partial dependence of the evolutionary changes of an organism to another specie which it has a close ecological relationship? This is observed in viceroy and monarch butterflies.

A. codependence

B. coevolution

C. coexistence

D. correlation

126. What will happen if the white blood cells are deteriorating?

A. There will be no production of hormone to clot the blood

B. The harmful bacteria in the body will not be destroyed

C. Oxygen cells will not be transported

D. The nerve impulses will not be carried to the brain

127. Which of the following process maintains the carbon dioxide-oxygen cycle in humans?

A. respiration

B. fossil fuels

C. photosynthesis

D. decomposition

128. What is the name of a tall, billowing cloud that looks like cotton ?

A. Nimbus

B. Cumulus

C. Cirrus

D. Stratus

129. Condensation occurs on a cool night because the temperature has reached the:

- A. Boiling Point.
- B. Melting Point.
- C. Freezing Point.
- D. Dew point**

130. Ozone is important to life on earth because?

- A. It reflects electrons from the sun
- B. Plants need it for photosynthesis
- C. Animals need it to breathe
- D. It absorbs harmful ultraviolet radiation**

131. The name of the process that traps heat in the atmosphere by using carbon dioxide and water vapor is __.

- A. greenhouse effect.**
- B. oxygen cycle.
- C. nitrogen cycle.
- D. carbon effect

132. What is the closest star to the Earth ?

- A. Moon
- B. Sun**
- C. Alpha Centauri
- D. Polaris

133. Who was the first human to set foot on the moon ?

- A. Galileo Galilae
- B. Johann Kepler
- C. John Glenn
- D. Neil Armstrong**

134. What is the part of the Sun that can only be seen during its total eclipse ?

- A. Sunspots
- B. Photosphere
- C. Corona**
- D. Chromosphere

135. The asteroid belt is located between what two planets?

- A. Earth and Mars
- B. Mars and Jupiter**
- C. Jupiter and Saturn
- D. Venus and Earth

136. What is the main force that keeps the planets in motion?

- A. Gravity**
- B. Light
- C. Sound
- D. Fusion

137. Life forms that live on the bottom of the ocean are called:

A. Benthos

- B. Nekton
- C. Zooplankton
- D. Phytoplankton

138. What is an El Niño?

- A. A change in the Northern Hemisphere jet stream brought on by changes in the air and water temperatures around Alaska.
- B. It is a change in the direction of the Gulf Stream which causes greatly increased periods of cold throughout Europe and across Asia.
- C. It is an increase in the amount of nutrient-rich waters that takes place off the west coast of Africa.

D. It is a rise in the surface ocean temperature along with a decrease in nutrient-rich waters that takes place in the Pacific Ocean off the west coast of South America.

139. Which mineral below is also known as fool's gold ?

- A. Galena
- B. Pyrite**
- C. Mica
- D. Silicon

140. Which is the name of the mineral found in sand ?

- A. Galena
- B. Granite
- C. Hydrogen
- D. Silica**

For numbers 142-146, refer to the vignette that follows. Choose the letter of the best answer from the choices that follow.

The Bataan Nuclear Power Plant is a nuclear power plant based in Morong, Bataan, Philippines. It was commissioned by President Ferdinand Marcos and building commenced in 1976. The plant's construction was completed in 1984. When Marcos was overthrown and replaced by Corazon Aquino, the nuclear power plant project was scrapped. A major concern is that the nuclear plant is located in an area where earthquakes are likely to occur. To this day, the Bataan Nuclear Power Plant has remained unused. Up until April 2007, the Philippine government had to pay 155,000 dollars every day to pay for the loan that funded the power plant's construction.

141. The following are known advantages of the use of nuclear power EXCEPT?

- A. decreased release of greenhouse gases
- B. increased reliance on fossil fuels**
- C. astronomical amounts of energy generation
- D. decreased cost of energy expenditure

142. Which among the following represents the single most important deterrent for the use of nuclear power?

A. disposal of radioactive wastes

- B. possibility of nuclear accidents
- C. scarcity of radioactive elements
- D. time frame needed for formalities, planning and building of a new nuclear plant

143. Nuclear power relies on which of the following chemical reactions?

- A. nuclear fusion
- B. binary fission
- C. chemiosmosis
- D. nuclear fission**

144. Which among the following world events sparked international debate against nuclear power?

- A. Chernobyl accident**
- B. Minamata accident
- C. Haiti incident
- D. Ormoc tragedy

145. Which among the following is the single most important effect of radiation exposure on human health?

- A. hypertension
- B. cancer**
- C. heart disease
- D. skin, muscle and bone problems

For numbers 146-150, refer to the scientific case described below. Choose the best answer to the questions that follow.

Waste disposal is a huge challenge for the urban municipality of Metropolis. It's expensive and time consuming to collect millions of tons of trash every year, and there are risks to the environment no matter how waste disposal is done. The city government is presently determining which method of waste disposal is best suited for their municipality.

146. Which among the following is NOT a suitable option for waste disposal in this city?

- A. vaporization**
- B. landfills
- C. incineration
- D. recycling

147. The following are possible adverse effects of the use of sanitary landfills, EXCEPT?

- A. Pollution of local environment
- B. Fatal accidents (scavengers buried under waste piles)
- C. Decreased amount of disease vectors**
- D. Dust, odor and noise pollution

148. Which among the following the single most important deterrent on the use of incinerators for garbage disposal?

- A. Heavy metal toxicity
- B. Dioxin and furan emissions**
- C. Highly toxic ashes flying around incinerator site
- D. Cost of operations

149. The following are solid waste products which can be recycled, EXCEPT?

- A. Leftover food**
- B. PET bottles
- C. Beverage cans
- D. PVC materials

150. At the level of people's homes, which among the following interventions can best help in solid waste management?

- A. Backyard burning of trash
- B. Composting
- C. Four o'clock habit
- D. Trash sorting prior to disposal**

SOCIAL STUDIES (40 points)

151. Who was the President of the 1987 Constitutional Commission?

- A. Cecilia Munoz Palma**
- B. Napoleon G. Rama
- C. Ambrosio B. Padilla
- D. Ahmad Domocao Alonto

152. What is regarded by the State as a "primary social economic force?"

- A. trade
- B. education
- C. commerce
- D. labor**

153. According to Article III, Section 15 of the Constitution, the writ of habeas corpus may be suspended in times of rebellion or what?

- A. invasion**
- B. terrorism
- C. martial law
- D. war

154. The Lower Chamber of the Congress of the Philippines is known by what name?

- A. Senate
- B. House of Representatives**
- C. National Assembly
- D. Congressional Lower Chamber

155. A Member of either house of Congress may be expelled by their fellow Members. For a Member to be expelled, how much of the total number of Members of a house must concur with the expulsion?

- A. three-fourths

B. two-thirds

C. majority

D. nine-tenths

156. In the event that both the President and Vice President are removed from office, who will act as President in the Philippines?

A. Executive Secretary of the Cabinet

B. Speaker of the House of Representatives

C. President of the Senate

D. Chief Justice of the Supreme Court

157. Who founded the La Liga Filipina?

A. Andres Bonifacio

B. Jose Rizal - correct answer

C. Emilio Aguinaldo

D. Emilio Jacinto

158. Who designed the Rizal Monument?

A. Carlos Nicoli

B. Antonio Luna

C. David Antel

D. Richard Kissling

159. In 1851, the first commercial bank in the Philippines and the far east, was established. What is the present name of this bank?

A. Bank of the Philippine Islands

B. Banco de las islas Filipinas

C. Bangko Central ng Pilipinas

D. Far East bank

160. Who was known as Hermano Pule?

A. Apolinario de la Cruz

B. Juan Arellano

C. Aurelio Tolentino

D. Pedro Feced

161. Who was the chief advisor of Gen. Emilio Aguinaldo?

A. Pedro Paterno

B. Jose Rizal

C. Apolinario Mabini

D. Felipe Calderon

162. Who was called "The First Filipino Diplomat" ?

A. Miguel Cuaderno

B. Felipe Calderon

C. Felipe Agoncillo

D. Robert Cortez

163. Who was the leader of the Cavite Mutiny, the first labor strike recorded in the Philippines?

- A. **Francisco la Madrid**
- B. Macliing Dulag
- C. Rafael de Izquierdo
- D. Patricio Montojo

164. Who was the first president of the Katipunan?

- A. Deodato Arellano**
- B. Teodoro Agoncillo
- C. Juan Arellano
- D. Andres Bonifacio

165. Under normal circumstances, when the price of a commodity falls, the quantity demanded will _____.

- A. increase**
- B. remain unchanged
- C. decrease
- D. nobody knows

166. Since human wants are unlimited and available resources are limited, what will a rational consumer do?

- A. Be jealous of the neighbours
- B. Renounce all wants and lead the life of a hermit
- C. Satisfy the most urgent needs which can be met within his/her income levels**
- D. Be resigned to unhappiness

167. When a consumer goes to market to purchase a commodity, he or she will purchase the _____ priced commodity, provided the commodities on offer are identical in all other respects.

- A. Highest
- B. Lowest**
- C. Will toss a coin to determine which one to buy
- D. Say "eenie, meenie, miney, mo" to decide which to buy

168. A business firm tries to maximise its _____ ?

- A. profits**
- B. revenue
- C. sales
- D. job retrenchment

169. When the economic activity in an economy is on the decline, then it is called a _____ ?

- A. boom
- B. recession**
- C. deflation
- D. stagnation

170. What usually constitutes the main source of a government's income?

- A. Loans
- B. Grants from World Bank

C. Profits from business undertakings

D. Taxes and levies

171. Sociology is the study of which among the following?

A. People in a society

B. Interactions among people

C. Social institutions

D. All of the above

172. Which theory believes that people in society battle over scarce resources like money, power, and prestige?

A. Structural functionalism

B. Conflict

C. Symbolic interaction

D. Feminist

173. Which research method utilizes public records such as marriage and divorce certificates, police records, and birth and death certificates?

A. Survey research

B. Participant observation

C. Experiment

D. Secondary data analysis

174. A person who weighs 100 pounds steps onto a scale which indicates that they weigh 200 pounds. He/she gets off the scale, steps back on, and it still reads 200 pounds. That scale is considered:

A. Reliable but not valid

B. Valid but not reliable

C. Both reliable and valid

D. Neither reliable nor valid

175. Which Philippine law provides that "courses on the life, works and writings of Jose Rizal, particularly his novels Noli Me Tangere and El Filibusterismo, shall be included in the curricula of all schools, colleges and universities, public or private?

A. RA 1425

B. RA 1081

C. RA 1452

D. RA 1542

176. What was the moral story of "The Story of the Moth"?

A. listen to elders who have more experiences

B. courage to face challenges in life

C. dying for another is worth it if it is for that person to see the light

D. all of the above

177. What is the complete name of our national hero?

A. Jose Pascasio Rizal Mercado y Alonso Realonda

B. Jose Protacio Mercado Rizal y Realonda Alonzo

C. Jose Protasio Alonzo Realonda y Mercado Rizal

D. Jose Protasio Rizal Mercado y Alonso Realonda

178. What does ilustrado mean?
- A. members in society with education in Europe**
 - B. members in society with public education
 - C. members in society who have part in the government
 - D. all of the above
179. Knowledge gained prior to birth and recollected through reason is associated with what type of philosophy?
- A. Consequentialism
 - B. Rationalism**
 - C. Constructivism
 - D. Empiricism
180. What is the view that the universe operates like a machine according to the law of cause and effect?
- A. Machinism
 - B. Interactionism
 - C. Toolism
 - D. Mechanism**
181. Suppose that there is a purpose, goal or final cause to the universe and that it all contributes to achieving an outcome. What is the word for the study of these ends or goals?
- A. Theology
 - B. Theocracy
 - C. Theodicy
 - D. Teleology**
182. What is the moral theory holding that the value of an action resides in its utility or use for the production of pleasure or happiness?
- A. Positivism
 - B. Consequentialism
 - C. Utilitarianism**
 - D. Deontology
183. Renaissance is a French term derived from the Italian word Rinascimento which meant _____?
- A. redux
 - B. return
 - C. rebirth**
 - D. remembrance
184. Pope Julius II commissioned whom to paint the Sistine Chapel ceiling in 1508?
- A. Raphael
 - B. Pier Matteo d'Amelia
 - C. Botticelli
 - D. Michelangelo**

185. Which artist drew 'The Vitruvian Man'?
- A. Sandro Botticelli
 - B. Titian
 - C. Michelangelo
 - D. Leonardo da Vinci**
186. Who achieved the remarkable record of producing a number one bestselling novel for seven consecutive years (from 1994-2000)?
- A. Stephen King
 - B. John Grisham**
 - C. Tom Clancy
 - D. Danielle Steele
187. Suppose you see water "leap" unexpectedly from one position to another. Which explanation is most scientific (ie, is best supported by current knowledge)?
- A. Divine forces caused the water to move the way it did
 - B. Random forces caused the water molecules to move in the same direction simultaneously**
 - C. The water molecules decided to move
 - D. Telekinesis moved the water
188. Which statement follows from evolutionary theory?
- A. Human intelligence is an adaptation**
 - B. Humans, like their genes, are doomed to be selfish
 - C. Kindness and tolerance are futile
 - D. Nature is cold, impersonal, and bleak
189. Which is most genetically adaptive in the greatest number of situations?
- A. Possessiveness**
 - B. Undiscriminating generosity
 - C. Severe cruelty
 - D. Risk-taking behavior
190. Why do men and women generally differ in their basic reproductive strategies?
- A. Men have a more difficult time communicating
 - B. Men want many mates and women want a few
 - C. In general, women are more agile than men and men are stronger than women
 - D. Biologically, women invest more in their offspring than men do**

EDUCATIONAL TECHNOLOGY (10 points)

191. What will you do first if your printer is not receiving the print command?
- A. Replace the ink cartridge.
 - B. Realign the heads of the printer.
 - C. Reset the printing properties.

D. Check if the printer cable is properly connected to the computer.

192. If you opened a document from your folder and added a new paragraph to it, how would you save the changes that you made?

- A. Select "Save As" and select a new folder and file name for it.
- B. Just click the "Save" icon on the toolbar or press Ctrl + S.**
- C. Just close the program. It will save automatically.
- D. Any of the above would work.

193. When would you use the "Save As" function?

- A. When you are saving a document for the first time and you need to choose a folder for the document and create a file name for it.**
- B. Every time you want to save something.
- C. Only when working with a Microsoft Word document.
- D. Only when you want to burn the file onto a CD.

194. If you accidentally deleted some text from a document, and you want it back, you should:

- A. Select "Redo" or press Ctrl +Y.
- B. Immediately close the document and reopen it.
- C. Restart the computer.
- D. Select "Undo" or press Ctrl + Z.**

195. Why do you think you should log-off of your computer when you leave class?

- A. You should not log-off your computer when you leave class.
- B. To prevent it from getting a virus.
- C. If you don't log-off, anyone can go on that computer and access your folder and files.**
- D. It helps to make sure that your work gets saved.

196. The best way to move some text to another part an essay would be to:

- A. Delete the text and then re-type it.
- B. Highlight the text, and then drag it into place with the mouse.
- C. Highlight the text, and press the up or down arrows to move it.
- D. Highlight the text, right-click, and use the "Cut" and "Paste" functions.**

197. Why should you be cautious when downloading files using file-sharing used to download files, music and videos?

- A. It is illegal to download music using these programs due to copyright laws.
- B. These files may contain viruses which can harm you computer.
- C. You might have to pay money in damages if you are caught.
- D. All of the above are true.**

198. If you are doing research for an essay, and you use some information from a website in your essay, you must:

- A. Do nothing. You can use this material in your essay freely.
- B. List the website in a bibliography at the end of your essay.**
- C. Type the information in bold letters.

D. Type the information in italic letters.

199. Why is a website like Wikipedia NOT the best choice for researching information about a topic?

A. Wikipedia and other "wikis" allow anyone to type information on their pages, which may be inaccurate.

B. It is the best choice.

C. There is too much information on the site to sort through which complicates things.

D. There are not enough images on the site.

200. If you wanted to create an animated presentations, what would be the BEST program to use among the following choices?

A. Microsoft Excel

B. Microsoft Power Point

C. Microsoft Word

D. Microsoft Access